Prices, Technical Indicators and Available Information

Calculate each indicator from actual values, then ask which observation or trading condition it supports.

Issues	Questions addressed
1-9: Averages and momentum	Which prices SMA, EMA and MACD use; how their conditions differ
10-16: Size of price changes	Volatility, TR, ATR and stop-loss distance
17-25: Trading conditions	RSI, volume, crossovers and combined conditions
26-30: Timing and features	Which values are known at decision time
31-34: External data	EPS and GDP periods, releases and revisions

Issue 5 illustrates one assumed EMA-rule trade, filled at the next Close; it does not rank indicators or tune parameters. Part C develops the account. Optional FinTech extensions outside this four-part course: Week 6 adds costs and Week 7 evaluates the full results.

Which AAPL prices do we share?

AAPL is Apple's ticker. Use the same price sample throughout rather than changing samples between questions.

Predict: Which three dates belong in the three-price average on 1/4?

```
import numpy as np
import pandas as pd
import yfinance as yf
aapl = yf.Ticker("AAPL").history(start="2024-01-02",
    end="2024-07-01", interval="1d", auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
pd.DataFrame({'Observation date':close.index[:5],
    'Close (USD/share)':close.iloc[:5].map("{:.2f}".format).values})
```

Observation date	Close (USD/share)
2024-01-02	185.64
2024-01-03	184.25
2024-01-04	181.91
2024-01-05	181.18
2024-01-08	185.56

The 1/2, 3, 4 Closes, 185.64, 184.25, 181.91, are the three inputs in Issue 1. The main sample has 124 observations through 6/28; 7/1 is excluded.

The source Close is split-adjusted, not dividend-adjusted. This part's returns exclude dividends, fees and tax. The short windows, 12/26/9 and Wilder14 are declared teaching settings, not optimal parameters.

Run the code in order; later issues use close and other objects created earlier. Figures use the same data; plotting code is hidden. Reproduction uses retained prices; a new download may include provider revisions.

ISSUE 01

What does an SMA of three Closes show?

A simple moving average (SMA) averages recent prices for comparison; it does not predict a future price.

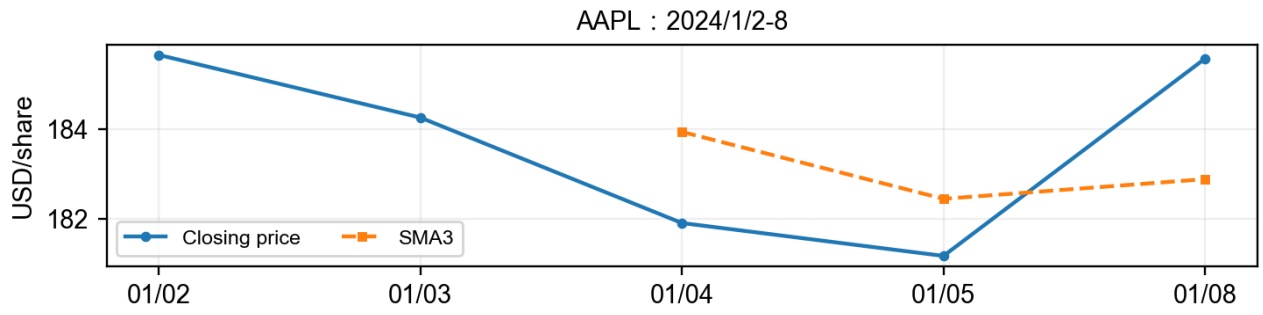
C is Close (USD/share). SMA3 uses today's and the previous two Closes. The first complete window is on 1/4; missing averages on 1/2, 3 must not be filled with zero.

$$\mathrm{SMA3}_t = \frac{C_t + C_{t-1} + C_{t-2}}{3}, \qquad \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333$$

Predict: On which date, 1/5 or 1/8, does Close cross from below to above the moving average?

```
sma3 = close.rolling(3, min_periods=3).mean()
dates = ["2024-01-04","2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    "SMA3":sma3.loc[dates].round(4)})
```

Observation date	Closing price	SMA3
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833



The 1/5 Close, 181.18, is below SMA3 182.4467; on 1/8, 185.56 is above 182.8833: an upward crossover.

An upward price crossover of the SMA is a common entry condition. The data are available after the close; meeting the condition does not mean a trade has filled.

Discussion

Which dates supply the prices for SMA3 on 1/5?

Sources: [3], [10].

ISSUE 02

Why can the moving average rise when Close falls?

A moving average changes with the entering and leaving prices, not just yesterday's and today's prices.

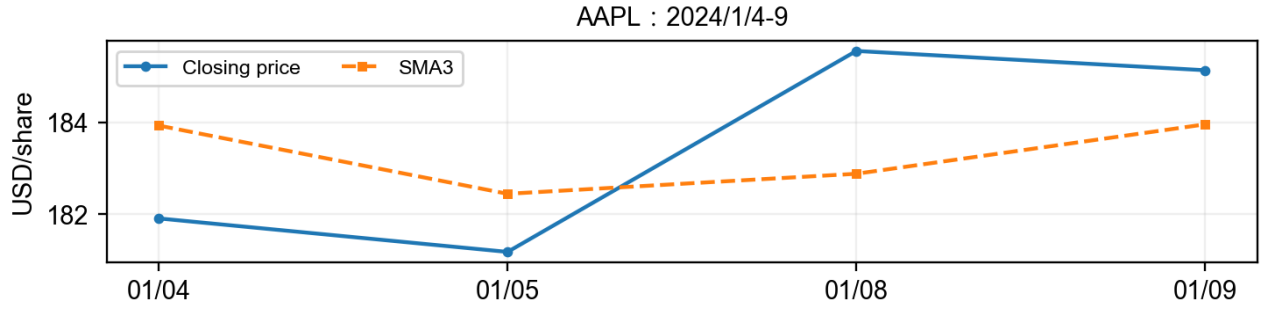
C is Close (USD/share). The 1/8 window uses 1/4, 5, 8; on 1/9 it uses 1/5, 8, 9. The shared 1/5 and 1/8 terms cancel.

$$\mathrm{SMA3}_9 - \mathrm{SMA3}_8 = \frac{C_9 - C_4}{3} \approx \frac{185.14 - 181.91}{3} \approx 1.0767$$

Predict: Close on 1/9 is 0.42 USD below 1/8. Must SMA3 also fall?

```
dates = ["2024-01-04","2024-01-05","2024-01-08","2024-01-09"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    "SMA3":sma3.loc[dates].round(4)})
```

Observation date	Closing price	SMA3
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833
2024-01-09	185.14	183.96



Close falls by 0.42 on 1/9, but adding 185.14 and removing 181.91 raises SMA3 by 1.0767 USD/share.

A rule that considers holding only when the SMA rises qualifies on 1/9. It differs from requiring a same-day price increase.

Discussion

Which entering and leaving prices make the moving average rise on 1/9?

Sources: [3], [10].

ISSUE 03

How does EMA combine today's price with yesterday's average?

An exponential moving average (EMA) retains part of the previous average and adds today's Close.

C and E are Close and EMA (USD/share); α weights today's price. With span=3, α=2/(3+1)=0.5. Initialize E with the 1/2 Close, 185.64, then update it from the previous E each day:

$$E_t = \alpha C_t + (1-\alpha)E_{t-1}, \qquad \alpha = \frac{2}{\text{span}+1}$$

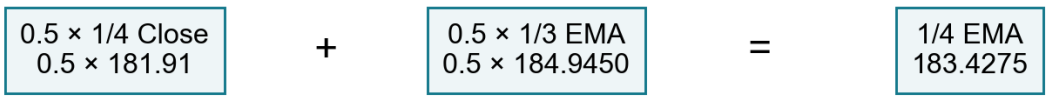
$$E_{\text{Jan3}} = 0.5(184.25) + 0.5(185.64) = 184.945$$

$$E_{\text{Jan4}} = 0.5(181.91) + 0.5(184.945) = 183.4275$$

Predict: Will the 1/4 EMA lie between 181.91 and 184.945?

```
ema3 = close.ewm(span=3, adjust=False).mean()
dates = ["2024-01-02","2024-01-03","2024-01-04"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    "EMA3":ema3.loc[dates].round(4)})
```

Observation date	Closing price	EMA3
2024-01-02	185.64	185.64
2024-01-03	184.25	184.945
2024-01-04	181.91	183.4275



Add the weighted amounts; both are USD/share

The 1/3 value is 184.945; use it as the previous average on 1/4. adjust=False uses this recursion.

Discussion

How much lower is EMA on 1/4 than on 1/3? Use the update equation: below which value must today's Close fall for EMA3 to decrease?

Sources: [4], [11].

ISSUE 04

Why does EMA3 retain prices older than three observations?

EMA is not limited to the last three prices. Older prices gradually lose influence; they do not disappear at the fourth observation.

Keep α=0.5 and first-price initialization. Substitute the 1/3 update into 1/4; E is EMA and C is Close (USD/share):

$$E_{\text{Jan4}} = 0.5C_{\text{Jan4}} + 0.25C_{\text{Jan3}} + 0.25C_{\text{Jan2}}$$

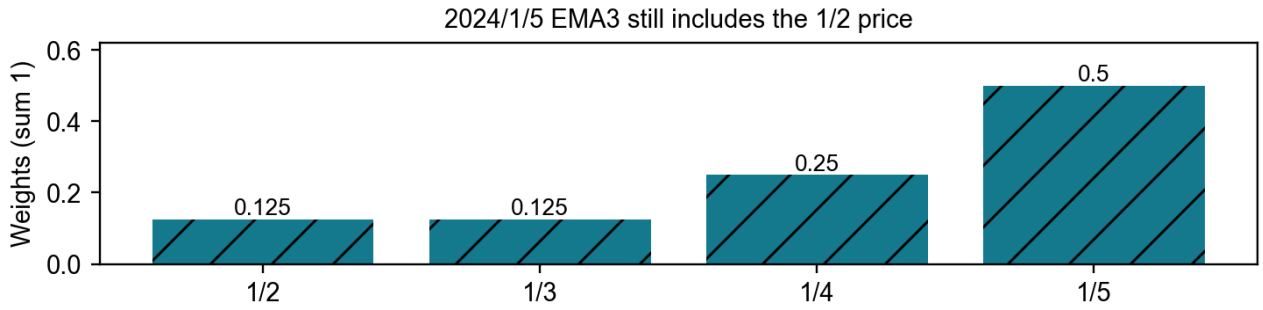
$$E_{\text{Jan5}} = 0.5C_{\text{Jan5}} + 0.25C_{\text{Jan4}} + 0.125C_{\text{Jan3}} + 0.125C_{\text{Jan2}}$$

$$E_{\text{Jan5}} \approx 0.5(181.18) + 0.25(181.91) + 0.125(184.25) + 0.125(185.64) \approx 182.304$$

Predict: The four weights sum to 1. Will their weighted prices reproduce EMA on 1/5?

```
dates = ["2024-01-02","2024-01-03","2024-01-04","2024-01-05"]
weights = np.array([0.125,0.125,0.25,0.5])
print(f"Four-price weighted sum:{(close.loc[dates]*weights).sum():.4f}")
print(f"Recursive EMA:{ema3['2024-01-05']:.4f}")
```

Four-price weighted sum:182.3037 Recursive EMA:182.3037



The 1/2 price has left the three-observation SMA window but retains weight 0.125 in EMA. The four-price sum and recursion both give 182.3037 USD/share.

EMA retains earlier history and gives greater weight to recent prices. This does not establish predictive ability. Different initialization can produce a different EMA.

Discussion

In EMA3 on 1/5, both 1/2 and 1/3 have weight 0.125. If initialized with the previous trading day's 2023/12/29 Close, what would the 1/2 weight be? Why are the two weights equal here?

Sources: [3], [4], [11].

ISSUE 05

Holding when EMA rises: what does the first trade earn?

Require stock when EMA3 rises, otherwise cash. Decide after the close and assume execution at the next trading day's Close.

Start in cash on 1/2; assume buying and selling one share at Close, without fees, dividends or interest. Find the first EMA rise and the first subsequent decline. Seeing today's rise does not earn today's return.

ema_after_close records stock after execution: True for stock, False for cash. entries marks cash-to-stock buy dates; exits marks stock-to-cash sell dates.

Predict: The 1/8 EMA rise creates a holding requirement. Can the same-day 185.56 be the buy price?

```
ema_up = ema3.diff() > 0
ema_after_close = ema_up.shift(1, fill_value=False)
entries = ema_after_close & ~ema_after_close.shift(1, fill_value=False)
exits = ~ema_after_close & ema_after_close.shift(1, fill_value=False)
ema_buy_date = close.index[entries][0]
ema_sell_date = close.index[exits & (close.index > ema_buy_date)][0]
ema_trade_return = close[ema_sell_date]/close[ema_buy_date]-1
print(f"Buy:{ema_buy_date} at {close[ema_buy_date]:.2f} USD/share")
print(f"Sell:{ema_sell_date} at {close[ema_sell_date]:.2f} USD/share")
print(f"Single-trade price return:{ema_trade_return:+.4%}")
```

Buy:2024-01-09 at 185.14 USD/share Sell:2024-01-17 at 182.68 USD/share Single-trade price return:-1.3287%

EMA rises from 182.3037 to 183.9319 on 1/8; buy at the 1/9 Close, 185.14. On 1/16 it falls from the previous trading day 1/12 value, 185.6982, to 184.6641: the first decline after buying. Sell at the 1/17 Close, 182.68. The holding period is 1/9 to 1/17.

$$\frac{182.68}{185.14} - 1 \approx -1.3287\%$$



An indicator rise only supplies an entry condition. This assumed trade at actual prices loses money; it cannot establish the whole strategy's results.

Discussion

If the 1/8 Close, 185.56, is used as the buy price, which execution assumption changes?

Sources: [4], [11].

ISSUE 06

Does negative MACD mean the price fell that day?

MACD compares shorter- and longer-span EMAs, not today's and yesterday's Closes.

MACD means Moving Average Convergence Divergence. E(12) and E(26) are span12, 26 EMAs, using Issue 3's 1/2 first-price initialization and update. Today's Close weights are α=2/13≈0.1538, 2/27≈0.0741; EMA12 gives recent Closes more weight. Units: USD/share.

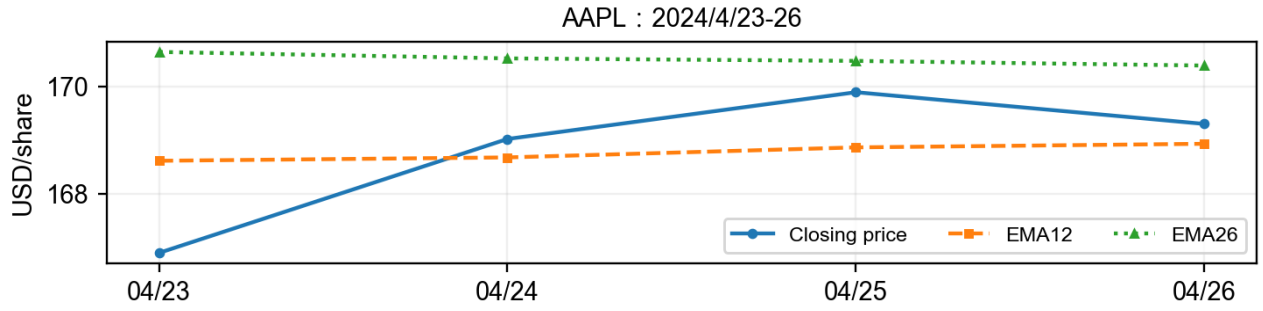
$$\mathrm{MACD}_t = E_t^{(12)} - E_t^{(26)}$$

$$\mathrm{MACD}_{\mathrm{Apr25}} \approx 168.8635 - 170.4726 \approx -1.609$$

Predict: The two EMAs have a negative difference on 4/25. Must that day's Close also fall?

```
ema12 = close.ewm(span=12, adjust=False).mean()
ema26 = close.ewm(span=26, adjust=False).mean()
macd = ema12 - ema26
dates = ["2024-04-24","2024-04-25","2024-04-26"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    "EMA12":ema12.loc[dates].round(5),"EMA26":ema26.loc[dates].round(5),
    "MACD":macd.loc[dates].round(5)})
```

Observation date	Closing price	EMA12	EMA26	MACD
2024-04-24	169.02	168.67681	170.51924	-1.84243
2024-04-25	169.89	168.86346	170.47263	-1.60917
2024-04-26	169.3	168.93062	170.38577	-1.45515



MACD is -1.6092 on 4/25, but Close rises from 169.02 on 4/24 to 169.89 USD/share, about +0.5147%. A lower short-span average does not mean a same-day price decline.

A common direction condition requires MACD above zero; 4/25 fails it. Asking whether MACD moves toward zero instead asks whether the gap is narrowing.

Discussion

Using the table, how much does MACD change from 4/24 to 4/26? How does each change in EMA12 and EMA26 contribute?

Sources: [4], [18].

ISSUE 07

Why can the histogram be positive when MACD is negative?

The histogram compares MACD with its own average. This differs from MACD's subtraction of two price EMAs.

S is MACD's 9-period EMA, the signal line, not an EMA of Close. As in Issue 3, α=2/(9+1)=0.2; initialize with the first MACD=0. The histogram value is Histogram in the equation. MACD, S and histogram units are USD/share.

$$S_t = 0.2\,\mathrm{MACD}_t + 0.8S_{t-1}, \qquad \mathrm{Histogram}_t = \mathrm{MACD}_t - S_t$$

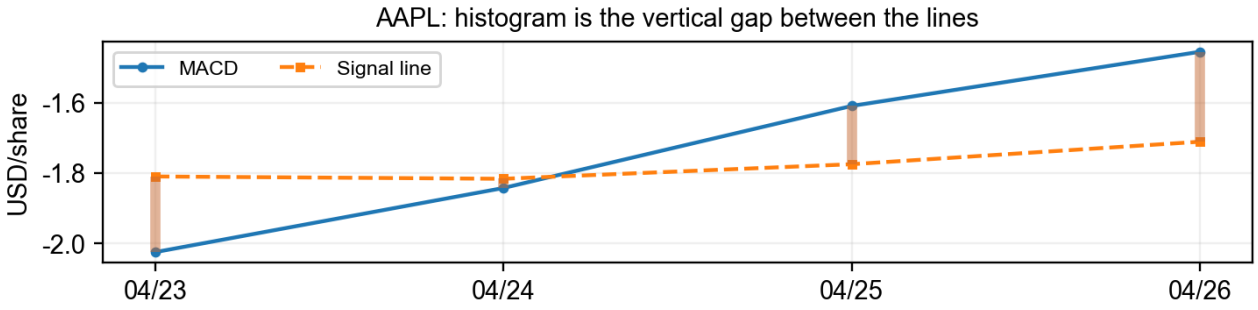
$$S_{\mathrm{Apr25}} \approx 0.2(-1.6092) + 0.8(-1.8163) \approx -1.775$$

$$\mathrm{Histogram}_{\mathrm{Apr25}} \approx -1.609 - (-1.775) = 0.166$$

Predict: Both lines are below zero. Can the histogram on 4/25 still be above zero?

```
signal_line = macd.ewm(span=9, adjust=False).mean()
histogram = macd - signal_line
dates = ["2024-04-24","2024-04-25"]
pd.DataFrame({'Observation date':dates,"MACD":macd.loc[dates].round(5),
    'Signal line':signal_line.loc[dates].round(5),
    'Histogram':histogram.loc[dates].round(5)})
```

Observation date	MACD	Signal line	Histogram
2024-04-24	-1.84243	-1.81625	-0.02618
2024-04-25	-1.60917	-1.77484	0.16566



The histogram is -0.02618 on 4/24 and +0.16566 on 4/25. MACD crosses above the signal line while remaining below zero.

A common entry condition is MACD crossing above its signal line. 4/25 meets it, but not MACD above zero; these are different signals.

Discussion

How much do MACD and the signal line each rise from 4/24 to 4/25? Use the S update to explain their different changes.

Sources: [4], [18], [23].

ISSUE 08

Are EMA12 below EMA26 and negative MACD two pieces of evidence?

The conditions are equivalent; requiring both adds no market information.

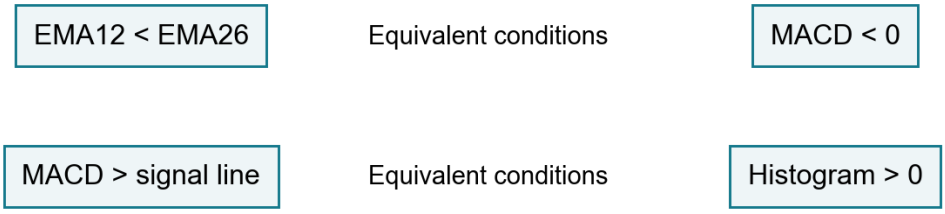
All three values use the same Closes, in USD/share. True means a condition holds that day; False means it does not. These are conditions, not trades or positions. The equivalence follows directly from MACD's definition, not from high correlation:

$$E^{(12)} < E^{(26)} \iff E^{(12)} - E^{(26)} < 0 \iff \text{MACD} < 0$$

Predict: Which two of the three condition columns must give the same True/False result?

```
dates = ["2024-04-24","2024-04-25"]
pd.DataFrame({'Observation date':dates,
    'EMA12 lower':(ema12.loc[dates] < ema26.loc[dates]).values,
    'MACD negative':(macd.loc[dates] < 0).values,
    'Histogram positive':(histogram.loc[dates] > 0).values})
```

Observation date	EMA12 lower	MACD negative	Histogram positive
2024-04-24	True	True	False
2024-04-25	True	True	True



The first two columns are True on 4/24 and 4/25. A positive histogram holds only on 4/25; it answers a different question.

Discussion

If both EMA12 below EMA26 and negative MACD are required, does removing the latter change the selected dates?

Sources: [18], [23].

ISSUE 09

Can short-horizon momentum be positive while longer-horizon momentum is negative?

A price rebound may still be below an earlier starting price. Different horizons can have different signs without contradiction.

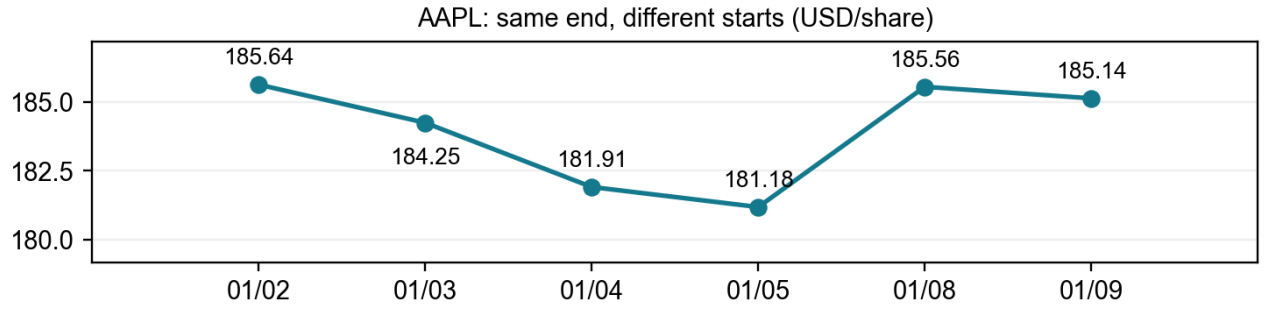
C is Close (USD/share), m is the rate of price change, and h counts intervals between observed trading days. Both horizons end on 1/9; only the starting point changes.

$$m_t^{(h)} = \frac{C_t}{C_{t-h}} - 1, \quad m_{\text{Jan9}}^{(3)} \approx \frac{185.14}{181.91} - 1 \approx 1.7756\%$$

Predict: 185.14 is above 181.91 but below 185.64. Will the two returns have the same sign?

```
momentum3 = close/close.shift(3)-1
momentum5 = close/close.shift(5)-1
starts = ["2024-01-04","2024-01-02"]
pd.DataFrame({'Start date':starts,'Start price':close.loc[starts].round(2).values,
    '1/9 end price':[round(close["2024-01-09"],2)]*2,
    'Return':[f"{momentum3['2024-01-09']:+.4%}",
            f"{momentum5['2024-01-09']:+.4%}"]})
```

Start date	Start price	1/9 end price	Return
2024-01-04	181.91	185.14	+1.7756%
2024-01-02	185.64	185.14	-0.2693%



The three-interval return is +1.7756%; the five-interval return is -0.2693%. Under a positive-momentum condition, changing the horizon changes the stock-holding condition.

Discussion

Five-interval momentum is negative on 1/9. For it to turn positive on the next trading day 1/10, which price must the 1/10 Close exceed? Why is the threshold not 185.64 on 1/2?

Sources: [5], [13].

ISSUE 10

Why is volatility nonzero when all three days fall?

Volatility measures each return's distance from the mean, not just its direction.

r is the decimal Close return; its mean is r with a bar, and sample standard deviation is σ with a hat. Close is 185.64 on 1/2 and 184.25, 181.91, 181.18 USD/share on 1/3, 4, 5.

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{3} (r_i - \bar{r})^2}{3-1}}$$

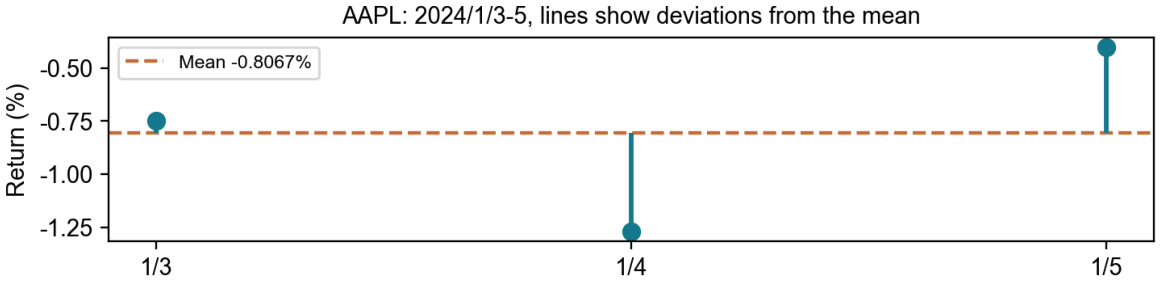
The three-return mean is about -0.0080669. Deviations are about +0.0005793, -0.0046332, +0.0040539; their sum is zero, so square them before summing. Divide by 2: because the three deviations sum to zero, only two can vary independently. Take the square root to return to return units.

$$\hat{\sigma} \approx \sqrt{\frac{0.0005793^2 + (-0.0046332)^2 + 0.0040539^2}{2}} \approx 0.004372 = 0.4372\%$$

Predict: All three returns are negative. Can their deviations from the mean have both signs?

```
daily_return = close.pct_change(fill_method=None)
first_returns = daily_return.iloc[1:4]
mean3 = first_returns.mean()
sd3 = first_returns.std(ddof=1)
pd.DataFrame({'Observation date':first_returns.index,
    'Return':first_returns.map("{:.4%}".format).values,
    'Deviation from mean':(first_returns-mean3).map("{:+.7f}".format).values})
```

Observation date	Return	Deviation from mean
2024-01-03	-0.7488%	+0.0005793
2024-01-04	-1.2700%	-0.0046332
2024-01-05	-0.4013%	+0.0040539



The three-return standard deviation is about 0.4372%. All returns have the same sign, but different deviations from the mean.

Discussion

Which day contributes the largest squared deviation, and approximately what fraction of the sum of squares?

Sources: [3], [14].

ISSUE 11

Does higher volatility mean a larger same-day loss?

A price rise can increase recent volatility. Volatility measures dispersion, not direction.

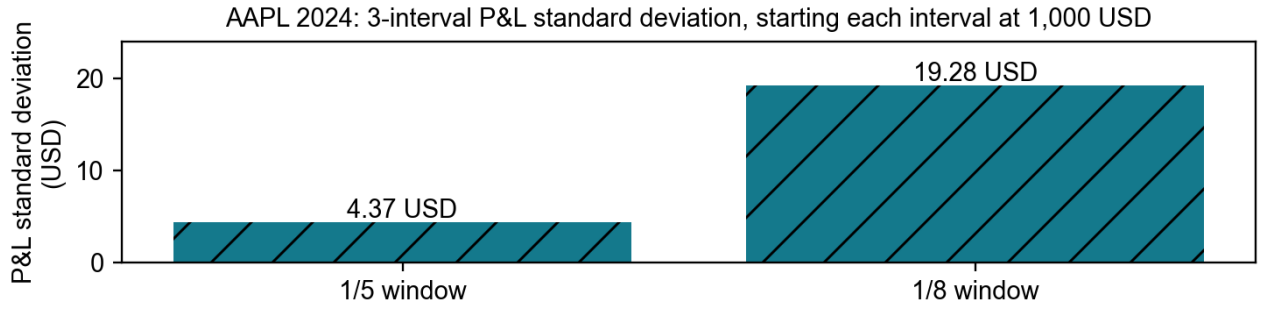
Compare 1/5 and 1/8 using three Close returns and sample standard deviation ddof=1; σ with a hat still denotes the return sample standard deviation. The 1/5 window is Issue 10's 1/3, 4, 5; on 1/8 it becomes 1/4, 5, 8, adding +2.4175% on 1/8 and removing -0.7488% on 1/3. Assume 1,000 USD starting value each interval: P&L is 1000 times the decimal return, with standard deviation 1000 times σ with a hat. The amounts are P&L standard deviations, not account values.

$$1000\hat{\sigma} \approx 1000(0.004372) \approx 4.37 \text{ USD}$$

Predict: The 1/8 price jumps 2.4175%. Can volatility be higher than on 1/5?

```
vol3 = daily_return.rolling(3, min_periods=3).std(ddof=1)
dates = ["2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,
    'Daily return':daily_return.loc[dates].map("{:+.4%}".format),
    'Three-interval volatility':vol3.loc[dates].map("{:.4%}".format),
    'P&L standard deviation (USD)':(1000*vol3.loc[dates]).map("{:.2f}".format)})
```

Observation date	Daily return	Three-interval volatility	P&L standard deviation (USD)
2024-01-05	-0.4013%	0.4372%	4.37
2024-01-08	+2.4175%	1.9278%	19.28



The 1/8 standard deviation, 1.9278%, corresponds to 19.28 USD, above 4.37 on 1/5. It is neither a same-day loss nor tomorrow's maximum loss.

Discussion

Can 19.28 USD be interpreted as tomorrow's maximum loss of 19.28 USD?

Sources: [3], [14], [26].

ISSUE 12

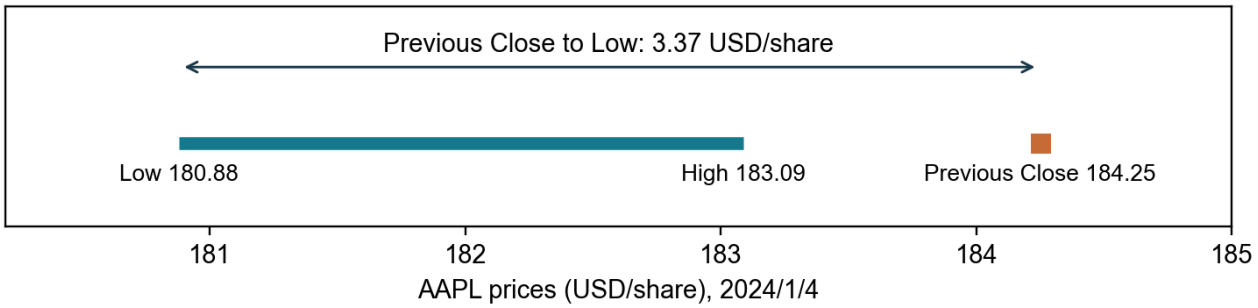
What decline does today's High-Low range miss?

Today's High and Low alone can understate the price change from yesterday's Close.

Assume one AAPL share is already held at the 1/3 Close: its value is 184.25 USD. On 1/4, High is 183.09 and Low is 180.88 USD; even High is below yesterday's Close. H, L denote today's High, Low; C is yesterday's Close. All differences below are USD/share.

$$H - L = 183.09 - 180.88 = 2.21, \qquad C - L = 184.25 - 180.88 = 3.37$$

Observation date	Observation	USD/share
2024/1/3	Closing price	184.25
2024/1/4	High	183.09
2024/1/4	Low	180.88



One share loses 3.37 USD from the previous Close 184.25 to the Low 180.88. The day's 2.21 high-low range does not cover the whole distance.

Today's High-Low is only 2.21 USD, but a share held since yesterday has lost 3.37 USD in value at today's Low. True Range (TR) therefore includes the previous Close. It measures magnitude, not future direction.

Discussion

Which omitted starting point explains the difference between 2.21 and 3.37?

Sources: [19], [21].

ISSUE 13

How can TR of 3.37 USD set a stop distance?

Calculate one day's TR, then use that magnitude for a later holding. It is not a buy signal.

H, L are today's High, Low; C is Close and t-1 is the previous trading day. Units: USD/share. Select the largest nonnegative distance; do not add the three:

$$\mathrm{TR}_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

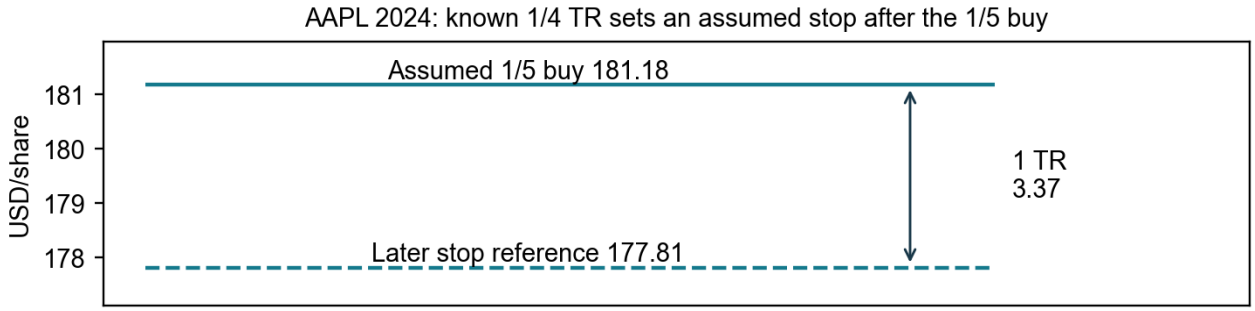
$$\mathrm{TR}_{\mathrm{Jan4}} = \max(2.21, |183.09 - 184.25|, |180.88 - 184.25|) = 3.37$$

TR is about 3.37 after the 1/4 close. Assume buying 10 shares at the actual 1/5 Close, 181.18, and fixing a later stop 1 TR below it: 181.18-3.37=177.81 USD/share. This is a teaching rule; common practice uses smoothed multi-day ATR instead.

Predict: If 10 shares are sold exactly at 177.81, is the loss 3.37 or 33.70 USD?

```
previous_close = close.shift(1)
distances = pd.concat([aapl.High-aapl.Low,
    (aapl.High-previous_close).abs(),
    (aapl.Low-previous_close).abs()],axis=1)
true_range = distances.iloc[1:].max(axis=1)
tr_distance = true_range["2024-01-04"]
tr_buy = close["2024-01-05"]
tr_stop = tr_buy-tr_distance
print(f"TR:{tr_distance:.2f}; trigger:{tr_stop:.2f} USD/share")
print(f"Loss for 10 shares filled exactly at trigger:{10*tr_distance:.2f} USD")
```

TR:3.37; trigger:177.81 USD/share Loss for 10 shares filled exactly at trigger:33.70 USD



The distance is 3.37 per share, or 33.70 USD for 10 shares. Assume a fall to 177.81 requests a sale. Triggering does not guarantee a fill price; gaps, slippage and fees can increase the loss.

Discussion

Keep the buy price and stop unchanged but buy 20 shares. If filled exactly at the trigger, excluding fees, approximately how many USD are lost?

Sources: [19], [21], [27].

ISSUE 14

TR measures a day; how does ATR combine days?

TR measures one day's magnitude. Average True Range (ATR) smooths daily TR values so a distance setting need not depend on a single day.

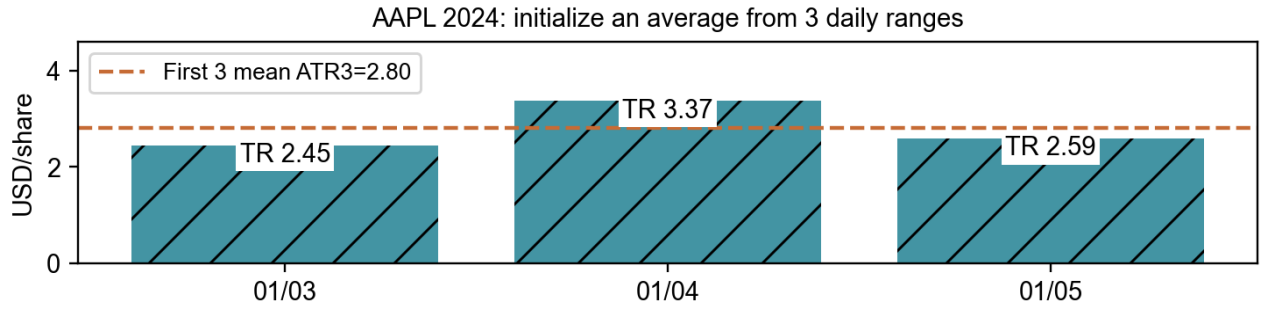
Use 3 periods for illustration; 14 periods is common. All units are USD/share. Initialize ATR with the mean of 3 TR values. Skip 1/2 because its previous Close is unavailable; the first 3 values are 1/3, 4, 5.

$$\mathrm{ATR3}_{\mathrm{Jan5}} = \frac{\mathrm{TR}_{\mathrm{Jan3}} + \mathrm{TR}_{\mathrm{Jan4}} + \mathrm{TR}_{\mathrm{Jan5}}}{3} \approx \frac{2.45 + 3.37 + 2.59}{3} \approx 2.80$$

Predict: Will the mean of 2.45, 3.37, 2.59 lie between their minimum and maximum?

```
seed_tr = true_range.iloc[:3]
atr_seed = seed_tr.mean()
pd.DataFrame({'Observation date':seed_tr.index,
    'TR (USD/share)':seed_tr.round(2).values})
```

Observation date	TR (USD/share)
2024-01-03	2.45
2024-01-04	3.37
2024-01-05	2.59



TR on 1/5 is 2.59. The first ATR3 is about 2.80 because it also includes 1/3 and 1/4, not just 1/5.

ATR14 starts with the first 14 valid TR values: 1/3 through 1/23 here, initially about 3.3507. From the 15th TR, use the next issue's Wilder update, not a freshly calculated simple moving average.

Discussion

On 1/5, TR is 2.59 and ATR3 is about 2.80. Which describes only that day, and which includes the previous two days?

Sources: [19], [21].

ISSUE 15

Today's TR is 4.42; why is ATR only 3.34?

ATR retains part of the previous average and adds today's range; it does not simply become today's TR.

Keep 3 periods: previous ATR is 2.8033 on 1/5; today's TR is 4.42 on 1/8, in USD/share. n is the period count and t denotes today. Wilder retains (n-1)/n of the old average and adds 1/n of today's TR:

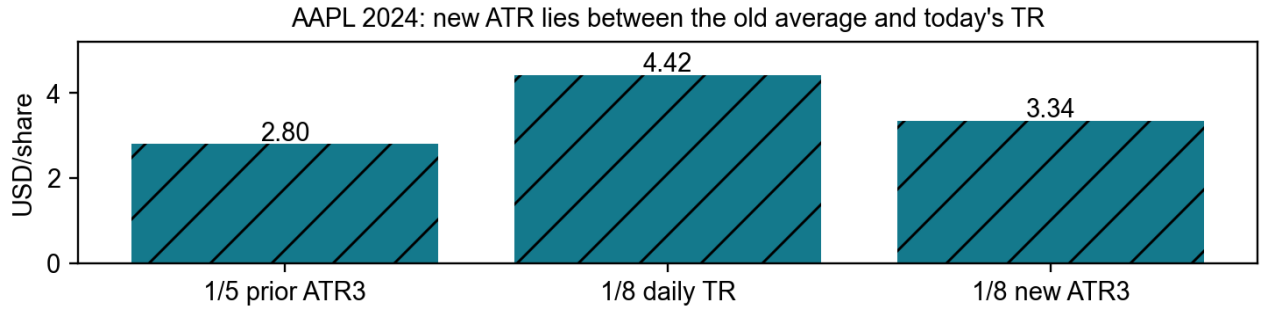
$$\mathrm{ATR}_t = \frac{(n-1)\mathrm{ATR}_{t-1} + \mathrm{TR}_t}{n}$$

$$\mathrm{ATR3}_{\mathrm{Jan8}} \approx \frac{2(2.8033) + 4.42}{3} \approx 3.3422$$

Predict: Today's TR4.42 exceeds prior ATR2.8033. Will the new ATR rise or fall?

```
def wilder_mean(values, n=14):
    result = pd.Series(np.nan,index=values.index)
    result.iloc[n-1] = values.iloc[:n].mean()
    for i in range(n,len(values)):
        result.iloc[i] = ((n-1)*result.iloc[i-1]+values.iloc[i])/n
    return result
atr3 = wilder_mean(true_range, n=3)
atr14 = wilder_mean(true_range)
dates = ["2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,"Today's TR":true_range.loc[dates].round(2),
    'Previous ATR':['None (initial-value row)',*atr3.shift(1).loc[dates[1:]].round(4)],
    "ATR3":atr3.loc[dates].round(4)})
```

Observation date	Today's TR	Previous ATR	ATR3
2024-01-05	2.59	None (initial-value row)	2.8033
2024-01-08	4.42	2.8033	3.3422



TR rises from 2.59 to 4.42, but ATR only rises from 2.80 to 3.34. A stop distance based on it is not determined solely by this day's range.

Do not repeatedly average the last 3 values. For 14 periods, use weights 13/14 and 1/14. The code also retains ATR14; later RSI uses the same smoothing function.

Discussion

If the next day's TR is below the current ATR3 of about 3.34, will updated ATR rise or fall?

Sources: [19], [21].

ISSUE 16

How do stops at 1 or 2 ATR differ?

Convert recent price variability into a stop distance. A larger ATR gives a wider distance at the same multiple.

Assume one share was bought at the 1/8 Close, 185.56 USD. ATR3 of about 3.3422 is calculated after the close to set a later stop, not to decide the already-completed buy. C is reference Close in USD/share; k is an assumed multiple, 1 or 2. Fees are excluded.

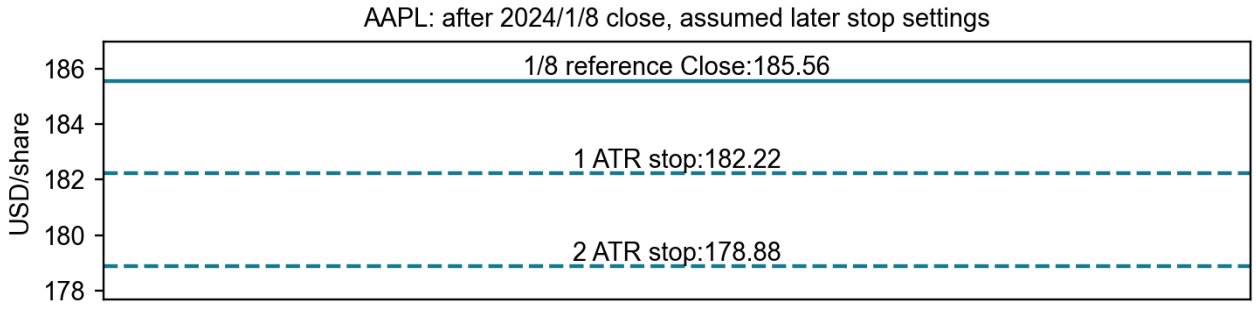
$$\text{Stop trigger} = C - k\,\text{ATR}, \quad 185.56 - 2(3.3422) \approx 178.88$$

Assume a price at or below the trigger activates a market sell order. Triggering does not guarantee execution at that price [27].

Predict: When the multiple changes from 1 to 2, is the trigger nearer to or farther from 185.56?

```
reference_close = close["2024-01-08"]
stop_atr = atr3["2024-01-08"]
pd.DataFrame({'Multiple':[1,2],
    'Distance (USD/share)':[stop_atr,2*stop_atr],
    'Trigger price':[reference_close-stop_atr,reference_close-2*stop_atr]}).round(2)
```

Multiple	Distance (USD/share)	Trigger price
1	3.34	182.22
2	6.68	178.88



The 1 ATR trigger is 182.22; 2 ATR gives 178.88. The latter allows a larger price change and planned loss. The multiples are not optimized; ATR is not a cap on tomorrow's decline.

Discussion

With the same buy price, 185.56, and fees excluded, what is the approximate one-share loss if filled exactly at each trigger?

Sources: [21], [27].

ISSUE 17

How does RSI separate daily gains and losses?

The Relative Strength Index (RSI) compares recent gains and losses. First split each Close change into two nonnegative values.

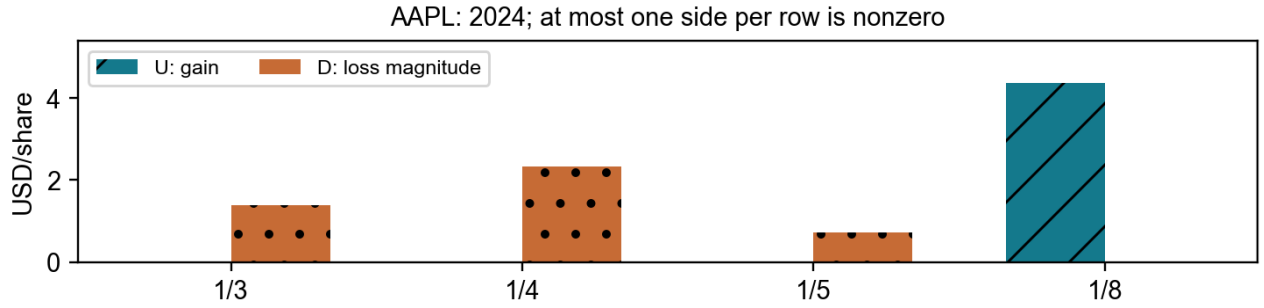
C is Close; ΔC is today minus previous Close. U is the gain and D the loss magnitude, in USD/share. Close is 185.64 on 1/2 and 184.25 on 1/3; record the 1.39 decline as D=1.39, not negative 1.39.

$$\Delta C_t = C_t - C_{t-1}, \quad U_t = \max(\Delta C_t, 0), \quad D_t = \max(-\Delta C_t, 0)$$

Predict: On 1/8 price rises from 181.18 to 185.56. What are U and D?

```
change = close.diff().iloc[1:]
gains, losses = change.clip(lower=0), (-change).clip(lower=0)
dates = ["2024-01-03","2024-01-04","2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates],
    'Change':change.loc[dates],"U":gains.loc[dates],
    "D":losses.loc[dates]}).round(2)
```

Observation date	Closing price	Change	U	D
2024-01-03	184.25	-1.39	0.0	1.39
2024-01-04	181.91	-2.34	0.0	2.34
2024-01-05	181.18	-0.73	0.0	0.73
2024-01-08	185.56	4.38	4.38	0.0



On 1/8, U=4.38 and D=0; on 1/3, U=0 and D=1.39. Both are 0 if Close is unchanged. Retain zeros when averaging in the next issue.

Discussion

For the four table rows from 1/3 to 1/8, what is total U minus total D? Which two Closes differ by this amount?

Sources: [20].

ISSUE 18

Why must RSI's average gain include days without gains?

Wilder14 initialization uses all 14 changes, including zeros; the denominator is not the count of rising or falling days.

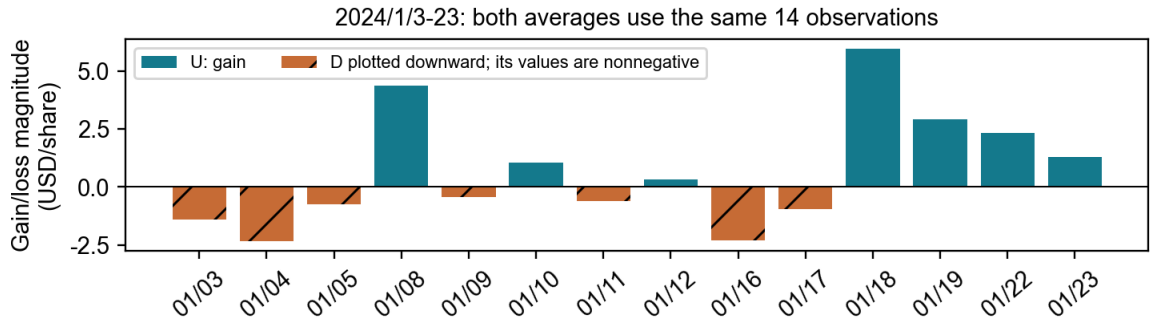
U and D come from the previous issue, in USD/share; bars denote smoothed averages. The first 14 changes span 1/3-23, with gains totaling 18.26 and loss magnitudes 8.72 USD/share. Divide each by 14 to initialize.

$$\bar{U}_{\text{Jan23}} = \frac{18.26}{14} \approx 1.3043, \quad \bar{D}_{\text{Jan23}} = \frac{8.72}{14} \approx 0.6229$$

Predict: If only U>0 days are retained, is this still the specified 14-observation mean?

```
gain_seed = gains.iloc[:14]
loss_seed = losses.iloc[:14]
initial_up, initial_down = gain_seed.mean(), loss_seed.mean()
print(f"14-observation U sum:{gain_seed.sum():.2f}; mean:{initial_up:.4f}")
print(f"14-observation D sum:{loss_seed.sum():.2f}; mean:{initial_down:.4f}")
```

14-observation U sum:18.26; mean:1.3043 14-observation D sum:8.72; mean:0.6229



Averaging only rising days changes the denominator; it does not reproduce this lesson's RSI14.

Discussion

Should a day with U=0 be removed from the 14-observation average gain?

Sources: [19], [20].

ISSUE 19

After a 0.68 decline, how do RSI's two averages update?

Use ATR's Wilder update separately for gains and loss magnitudes; today's decline is not a negative D.

Close is 195.18 on 1/23 and 194.50 USD/share on 1/24, so U=0 and D=0.68. Previous average gain is 1.3043 and average loss magnitude 0.6229 USD/share.

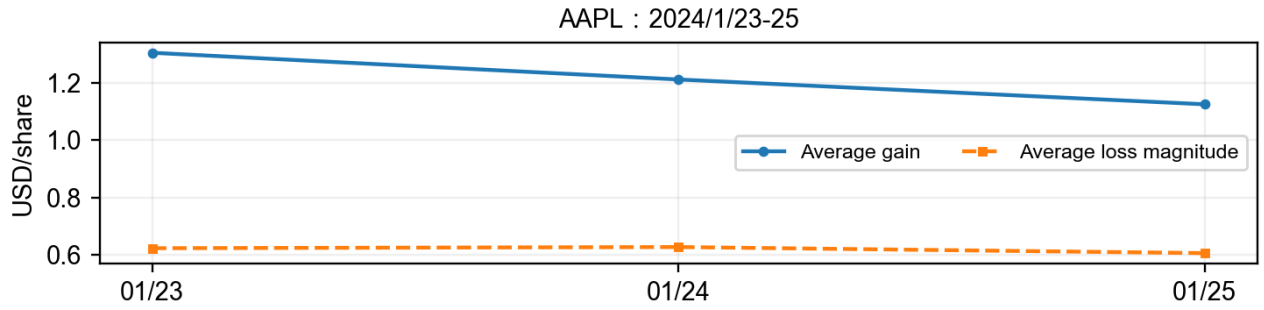
$$\bar{U}_t = \frac{13\bar{U}_{t-1} + U_t}{14}, \quad \bar{D}_t = \frac{13\bar{D}_{t-1} + D_t}{14}$$

$$\bar{U}_{24} \approx \frac{13(18.26/14) + 0}{14} \approx 1.2111, \quad \bar{D}_{24} \approx \frac{13(8.72/14) + 0.68}{14} \approx 0.6269$$

Predict: With no gain today, does average gain immediately become 0 or decline gradually?

```
up, down = wilder_mean(gains), wilder_mean(losses)
dates = ["2024-01-23","2024-01-24","2024-01-25"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    "U":gains.loc[dates].round(2),"D":losses.loc[dates].round(2),
    'Average gain':up.loc[dates].round(4),
    'Average loss magnitude':down.loc[dates].round(4)})
```

Observation date	Closing price	U	D	Average gain	Average loss magnitude
2024-01-23	195.18	1.29	0.0	1.3043	0.6229
2024-01-24	194.5	0.0	0.68	1.2111	0.6269
2024-01-25	194.17	0.0	0.33	1.1246	0.6057



The 1/23 row is Issue 18's 14-observation initialization. On 1/24 average gain falls to 1.2111 and average loss magnitude rises to 0.6269. The 1/25 decline of 0.33 is below 0.6269, so average loss magnitude falls to 0.6057.

Discussion

Why is average gain still above zero on 1/24?

Sources: [19], [20].

ISSUE 20

Does RSI 65.89 mean a stock return of 65.89%?

RSI expresses average gain as a share of total magnitude on a 0-100 scale, not a holding return.

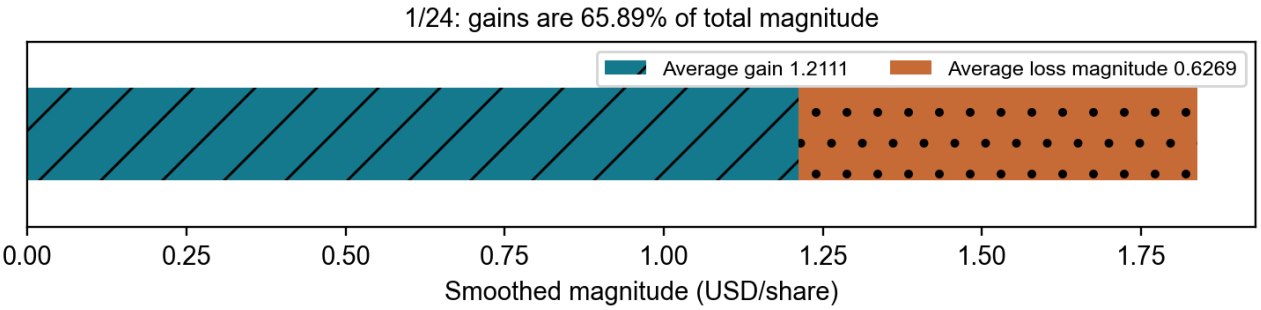
U and D with bars are Wilder-smoothed gain and loss magnitude in USD/share. Their sum is total magnitude; the currency units cancel in the ratio. This lesson leaves a zero-total case undefined.

$$\mathrm{RSI} = 100\frac{\bar{U}}{\bar{U}+\bar{D}}, \quad 100\frac{1.2111}{1.2111+0.6269} \approx 65.89$$

Predict: Price falls 0.3484% on 1/24. Can RSI still be above 50?

```
total = up + down
rsi14 = 100*up/total.where(total > 0)
dates = ["2024-01-23","2024-01-24","2024-01-25"]
pd.DataFrame({'Observation date':dates,"RSI14":rsi14.loc[dates].round(2),
    'Daily price return':daily_return.loc[dates].map("{:+.4%}".format)})
```

Observation date	RSI14	Daily price return
2024-01-23	67.68	+0.6653%
2024-01-24	65.89	-0.3484%
2024-01-25	64.99	-0.1697%



RSI above 50 means smoothed gains exceed loss magnitudes. It does not mean a same-day rise or a 65.89% profit.

For average loss magnitude above 0, source [20] writes RSI=100-100/(1+RS), with RS=average gain÷average loss magnitude. On 1/24: RS=1.2111÷0.6269≈1.932; 100-100/2.932≈65.89, equivalent to this page's formula.

Discussion

What is RSI if average loss magnitude is 0 and average gain is above 0?

Sources: [20].

ISSUE 21

Does RSI above 70 guarantee a fall next trading day?

70 is a common overbought threshold, not a guarantee of reversal. Check all qualifying dates in this sample, not just one counterexample.

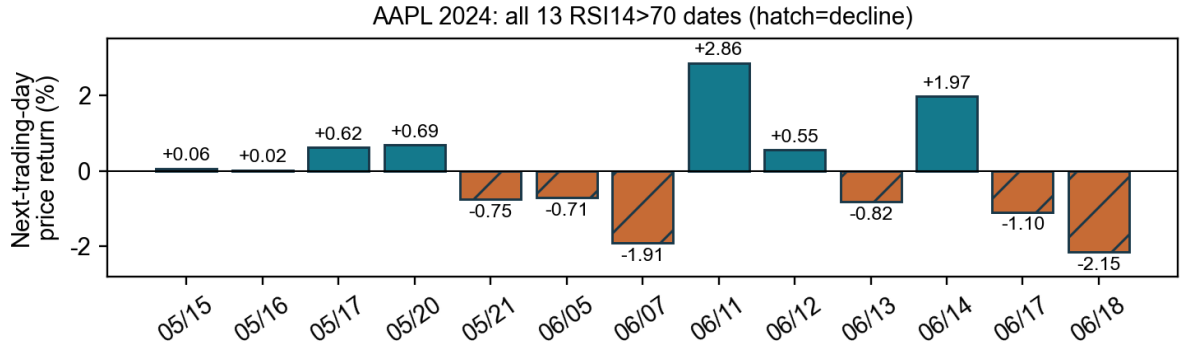
AAPL, 2024/1/2-6/28; use RSI14 from Issues 18-20. Include only defined RSI with a next Close; do not fill with 0. C is Close (USD/share), and t+1 the next trading day. Close is 207.15 on 6/11 and 213.07 on 6/12.

$$r = \frac{C_{t+1}}{C_t} - 1; \qquad \frac{213.07}{207.15} - 1 \approx +2.8578\%$$

Predict: Is every next-trading-day return after RSI>70 in this sample negative?

```
events = pd.DataFrame({"RSI":rsi14.reindex(close.index),
    'Closing price':close, 'Next trading date':pd.Series(close.index,index=close.index).shift(-1),
    'Next Close':close.shift(-1)}).dropna()
events['Next-day return'] = events['Next Close']/events['Closing price']-1
high_rsi = events.loc[events["RSI"] > 70]
shown = high_rsi.loc[["2024-05-15","2024-06-07","2024-06-11","2024-06-18"]].copy()
shown.insert(0,'Observation date',shown.index)
shown[["RSI",'Closing price','Next Close']] = shown[["RSI",'Closing price','Next Close']].round(2)
shown['Next-day return'] = shown['Next-day return'].map("{:+.4%}".format)
shown
```

Observation date	RSI	Closing price	Next trading date	Next Close	Next-day return
2024-05-15	70.83	189.72	2024-05-16	189.84	+0.0632%
2024-06-07	70.92	196.89	2024-06-10	193.12	-1.9148%
2024-06-11	75.22	207.15	2024-06-12	213.07	+2.8578%
2024-06-18	73.57	214.29	2024-06-20	209.68	-2.1513%



Of 13 events, 7 rise and 6 fall: 6/11→12 rises 2.8578%, and 6/7→10 falls 1.9148%. The trading day after 6/18 is 6/20. Consecutive high-RSI dates are not independent trials; 7/13 is this sample's proportion, not a future rise probability.

A no-new-buy rule for RSI>70 excludes all 13 events: it may avoid declines but also miss further rises. Signals form after the close; the plotted changes are not returns obtainable by a later order. See Issue 5 for execution timing.

Discussion

Compare 6/7 and 6/11: why does RSI>70 not directly imply next-trading-day return<0?

Sources: [20].

ISSUE 22

Why can price fall when volume rises 58%?

A volume change concerns shares traded, not a holding return; the two are not interchangeable.

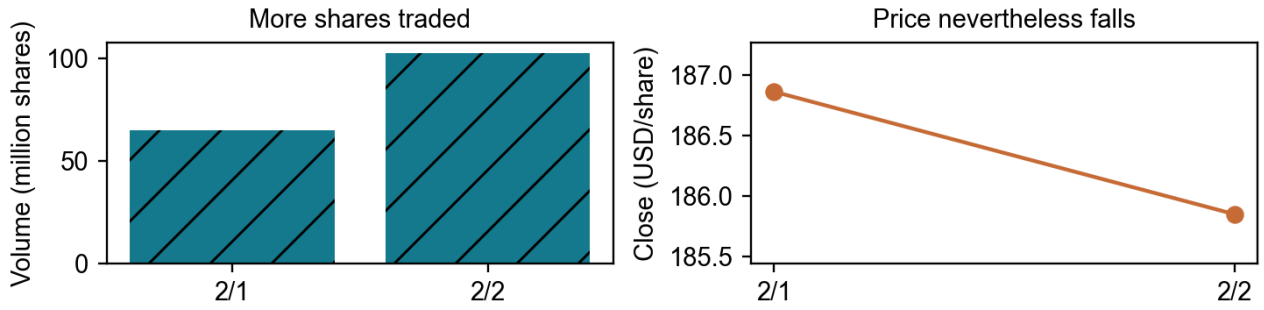
V is daily volume (shares), C is Close (USD/share), g is volume growth, and r is price return. Both compare 2/2 with the previous trading day 2/1, using their 2/1 values as denominators.

$$g = \frac{V_2}{V_1} - 1 = \frac{102551700}{64885400} - 1 \approx 58.05\%, \quad r \approx \frac{185.85}{186.86} - 1 \approx -0.54\%$$

Predict: Does more trading on 2/2 mean shareholders earn 58.05%?

```
volume = aapl["Volume"]
volume_change = volume/volume.shift(1)-1
dates = ["2024-02-01","2024-02-02"]
pd.DataFrame({'Observation date':dates,'Closing price':close.loc[dates].round(2),
    'Volume (shares)':volume.loc[dates].values,
    'Price return':["-",f"{daily_return['2024-02-02']:+.2%}"],
    'Volume change':["-",f"{volume_change['2024-02-02']:+.2%}"]})
```

Observation date	Closing price	Volume (shares)	Price return	Volume change
2024-02-01	186.86	64885400	-	-
2024-02-02	185.85	102551700	-0.54%	+58.05%



Volume on 2/2 is 102.55 million shares, 58.05% above 64.89 million; Close nevertheless falls 1.01 USD.

Higher volume alone does not measure buyers' bullishness. Every trade has a buyer and seller; daily total volume does not identify the initiating side.

Discussion

Is the denominator of 58.05% yesterday's stock price or shares traded?

Sources: [15], [24].

ISSUE 23

Can volume exceed yesterday's but remain below its usual level?

Identify the comparison baseline. After a low-volume day, today's increase may still be below the previous 20-observation average.

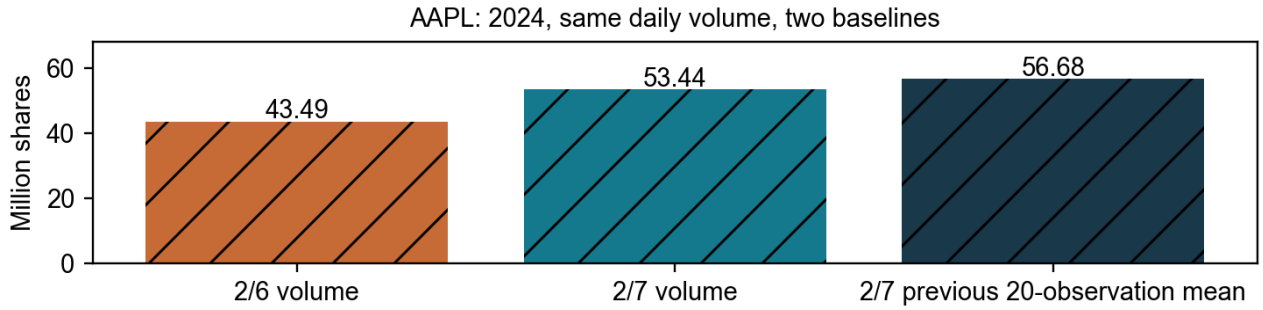
V is shares traded; M is average volume over the previous 20 observations, excluding today. On 2/7, M uses 20 trading days from 1/9 to 2/6: 1,133,676,500 shares÷20=56,683,825 shares. V/M is a multiple; 1 means average volume, not a 100% investment return.

$$\frac{V_{\text{Feb7}}}{M_{\text{Feb7}}} = \frac{53439000}{56683825} \approx 0.943, \quad \frac{53439000}{43490800} - 1 \approx 22.87\%$$

Predict: Are 53,439,000 shares on 2/7 above both yesterday's volume and the previous 20-observation mean?

```
prior20 = volume.shift(1).rolling(20,min_periods=20).mean()
volume_multiple = volume/prior20
dates = ["2024-02-06","2024-02-07"]
pd.DataFrame({'Observation date':dates,'Volume (shares)':volume.loc[dates],
    'Previous 20-observation mean volume':prior20.loc[dates].round(0),
    'Multiple':volume_multiple.loc[dates].round(3)})
```

Observation date	Volume (shares)	Previous 20-observation mean volume	Multiple
2024-02-06	43490800	57466510.0	0.757
2024-02-07	53439000	56683825.0	0.943



Volume on 2/7 rises 22.87% from yesterday but is only 0.943 times average volume. A rising-price condition with above-average volume is not met.

Close rises from 189.30 on 2/6 to 189.41 USD/share on 2/7, about 0.06%. The original condition excludes this date because volume is below its average.

Discussion

If higher volume instead means above yesterday's volume, does 2/7 qualify?

Sources: [3], [15], [24].

ISSUE 24

Does an average being above another mean a golden cross?

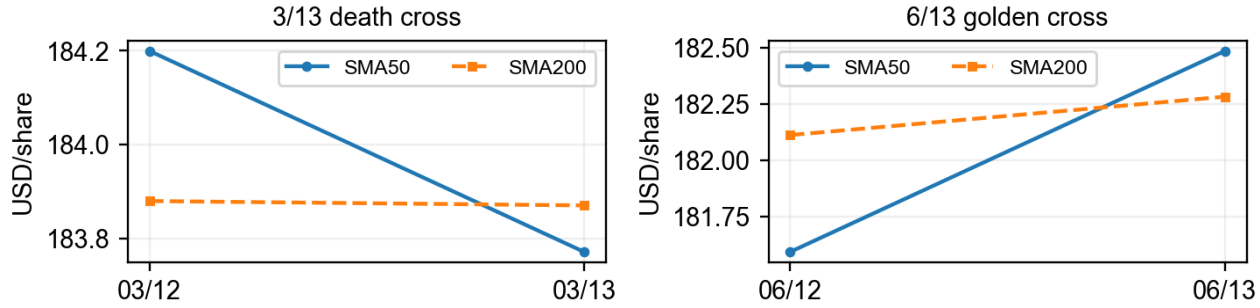
A crossover changes the ordering between two days; being above describes only today's state.

SMA50 and SMA200 average 50 and 200 Closes, in USD/share. A golden cross occurs when yesterday's 50 line is not above the 200 line, but today's is above; a death cross reverses this. Obtain earlier history to calculate 200 observations.

Predict: The 50 line is higher on 6/13. Which other day determines whether it just crossed?

```
long_history = yf.Ticker("AAPL").history(start="2023-01-01",
    end="2024-07-01",interval="1d",auto_adjust=False,actions=True)
long_history.index = long_history.index.strftime("%Y-%m-%d")
long_close = long_history["Close"]
assert np.allclose(long_close.loc[close.index],close,rtol=0,atol=1e-10)
sma50,sma200 = long_close.rolling(50).mean(),long_close.rolling(200).mean()
golden = (sma50>sma200) & (sma50.shift(1)<=sma200.shift(1))
death = (sma50<sma200) & (sma50.shift(1)>=sma200.shift(1))
dates=["2024-03-12","2024-03-13","2024-06-12","2024-06-13"]
pd.DataFrame({'Observation date':dates,"SMA50":sma50.loc[dates].round(4),
    "SMA200":sma200.loc[dates].round(4)})
```

Observation date	SMA50	SMA200
2024-03-12	184.1992	183.8796
2024-03-13	183.7712	183.8703
2024-06-12	181.5936	182.1121
2024-06-13	182.4854	182.2824



Death cross: 3/12 SMA50 184.1992>183.8796, then 3/13 183.7712<183.8703. Golden cross: 6/12 181.5936<182.1121, then 6/13 182.4854>182.2824. These are candidate conditions, not guarantees of subsequent price direction.

Discussion

After 6/13, while the 50 line stays above the 200 line, is every day a new golden cross?

Sources: [3], [24].

ISSUE 25

Which dates require stock when both MACD and RSI must qualify?

Explicit indicator conditions can be calculated and checked more directly than a vague claim of a favorable trend.

Here stock is required only if histogram>0 and RSI14<70; otherwise cash. AND requires both; OR requires at least one. In the AND column, True requires stock after that day's close; False means at least one condition fails. These are conditions, not fills. Incomplete inputs remain nan, not False or zero.

Predict: On 6/11, histogram 0.5214>0 but RSI is 75.22. Does AND hold?

```
dates = ["2024-05-06","2024-06-07","2024-06-10","2024-06-11"]
valid = histogram.notna() & rsi14.reindex(close.index).notna()
combined = ((histogram>0) & (rsi14.reindex(close.index)<70)).where(valid)
pd.DataFrame({'Observation date':dates,'Histogram':histogram.loc[dates].round(4),
    "RSI":rsi14.loc[dates].round(2),
    'AND holds':combined.loc[dates].values})
```

Observation date	Histogram	RSI	AND holds
2024-05-06	1.6806	63.79	True
2024-06-07	0.0507	70.92	False
2024-06-10	-0.1925	60.18	False
2024-06-11	0.5214	75.22	False

Date	Histogram>0	RSI<70	AND requirement
05/06	Met	Met	Stock
06/07	Met	Not met	Cash
06/10	Not met	Met	Cash
06/11	Met	Not met	Cash

Both hold on 5/6, requiring stock. RSI fails on 6/7 and 11; the histogram fails on 6/10.

This is only a candidate trading rule; execution and account amounts are not yet specified. ATR may supply distance, but not direction by itself.

Discussion

If only AND changes to OR on 6/11, does the stock requirement change?

Sources: [18], [20], [24].

ISSUE 26

Can today's final-Close SMA be calculated at 15:00?

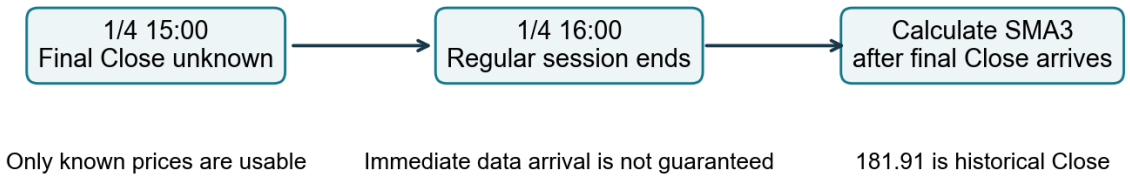
A price labeled with today's date is not necessarily known throughout that day.

Suppose SMA3 is needed at 15:00 New York time on 1/4 using three final Closes. Regular trading ends at 16:00; the previous two Closes have arrived, but final 1/4 Close does not yet exist.

Close date	Known at 15:00 (USD/share)	Usable for original rule
2024/1/2	185.64	Yes
2024/1/3	184.25	Yes
2024/1/4	Not yet known	No

$$\mathrm{SMA3}_{\mathrm{Jan4}} = \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333$$

181.91 is the 1/4 Close in the historical file; it cannot be backfilled as known at 15:00. Intraday prices can support another rule, but its changed inputs must be declared.



Correct arithmetic cannot replace unavailable inputs. A daily historical file does not establish the actual historical receipt time.

Discussion

If 184.25 from 1/3 is used as the 1/4 Close, is this still the original SMA3?

Sources: [1], [2].

ISSUE 27

Which return interval does the 1/4 feature predict?

A feature is a known input; the target is an outcome known later. Sharing a row does not make both known simultaneously.

C is Close in USD/share; y is the decimal price return to the next Close. t is the feature date; t+1 is the next trading day, not necessarily the next calendar day.

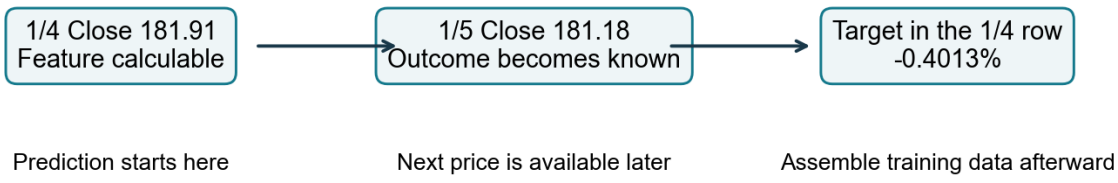
nan means the next Close is missing and the target cannot yet be calculated; it does not mean Close or return is 0.

$$y_t = \frac{C_{t+1}}{C_t} - 1, \qquad y_{\text{Jan4}} \approx \frac{181.18}{181.91} - 1 \approx -0.4013\%$$

Predict: The sample ends on 6/28. Is the next-interval target for 6/28 available?

```
target = close.shift(-1)/close-1
dates = ["2024-01-04","2024-01-05","2024-06-28"]
pd.DataFrame({'Feature date':dates,"Today's Close":close.loc[dates].round(2),
    'Following Close':close.shift(-1).loc[dates].round(2),
    'Target':target.loc[dates].map(lambda x:'Unknown' if pd.isna(x) else f"{x:+.4%}")})
```

Feature date	Today's Close	Following Close	Target
2024-01-04	181.91	181.18	-0.4013%
2024-01-05	181.18	185.56	+2.4175%
2024-06-28	210.62	nan	Unknown



shift(-1) only aligns data; it does not make the 1/5 price known on 1/4 or imply exposure to that return interval.

Discussion

Using y from the 1/4 row as a 1/4 input uses which not-yet-observed day's price?

Sources: [5], [6].

ISSUE 28

Why is a closer centered average unavailable that day?

It includes the following day's price under today's label. A closer line does not mean earlier information.

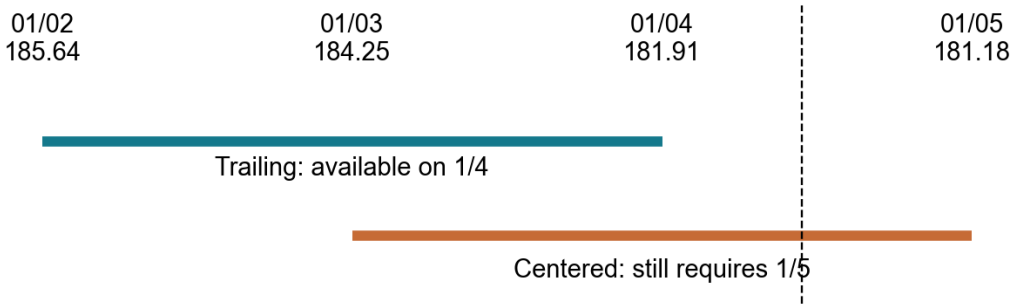
The trailing SMA on 1/4 uses 1/2, 3, 4; the centered mean uses 1/3, 4, 5. C is Close in USD/share; both divide by 3.

$$\text{Centered}_{\text{Jan4}} = \frac{184.25 + 181.91 + 181.18}{3} \approx 182.4467$$

Predict: Without prices from 1/5 onward, can the centered mean on 1/4 be calculated?

```
centered3 = close.rolling(3,center=True,min_periods=3).mean()
prefix = close.loc[:"2024-01-04"]
print(f"1/4 trailing:{sma3['2024-01-04']:.4f}")
print(f"Retrospective 1/4 centered:{centered3['2024-01-04']:.4f}")
print('Centered as of 1/4:',prefix.rolling(3,center=True).mean().iloc[-1])
```

1/4 trailing:183.9333 Retrospective 1/4 centered:182.4467 Centered as of 1/4: nan



Trailing 183.9333 is calculable; centered 182.4467 must wait for 1/5. nan means that as of 1/4, the 1/5 price is missing, so the centered value is unavailable.

Discussion

Does the centered value being closer to the 1/4 Close, 181.91, establish predictive ability?

Sources: [3], [6].

ISSUE 29

How far is momentum 0.4383% from the earlier mean?

Subtract the earlier mean and divide by the earlier standard deviation to express distance without units.

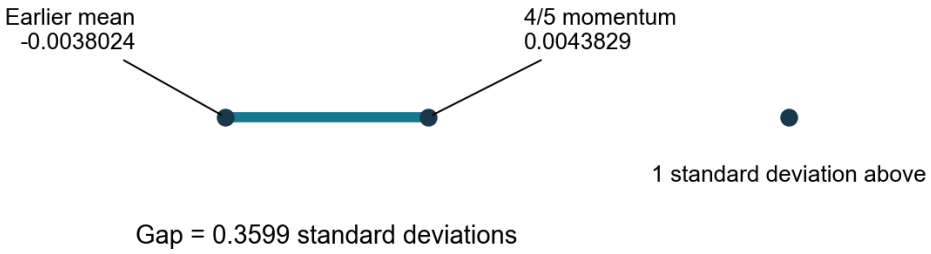
m is the three-interval momentum from Issue 9; μ and σ̂ are the mean and sample standard deviation of the 62 valid momentum values through 4/4. z is standardized, not a return percentage.

$$z = \frac{m - \mu}{\hat{\sigma}}, \quad z_{\text{Apr5}} \approx \frac{0.0043829 - (-0.0038024)}{0.0227433} \approx 0.3599$$

Predict: The 4/5 momentum exceeds the negative earlier mean. Will z be positive or negative?

```
fit = momentum3.loc[:"2024-04-04"].dropna()
mu, scale = fit.mean(),fit.std(ddof=1)
apr5 = momentum3["2024-04-05"]
z_earlier = (apr5-mu)/scale
print(f"Observations:{len(fit)}; mean{mu:.7f}; standard deviation{scale:.7f}")
print(f"4/5 momentum{apr5:.7f}; standardized{z_earlier:.4f}")
```

Observations:62; mean-0.0038024; standard deviation0.0227433 4/5 momentum0.0043829; standardized0.3599



0.3599 means about 0.36 standard deviations above the earlier mean, not a 0.3599% profit.

A model may use z to compare inputs on different scales. This conversion requires a finite, positive denominator.

Discussion

Keeping this page's μ and σ̂, what minimum momentum percentage gives z=1?

Sources: [6], [14].

ISSUE 30

Why does the same 4/5 momentum yield another z?

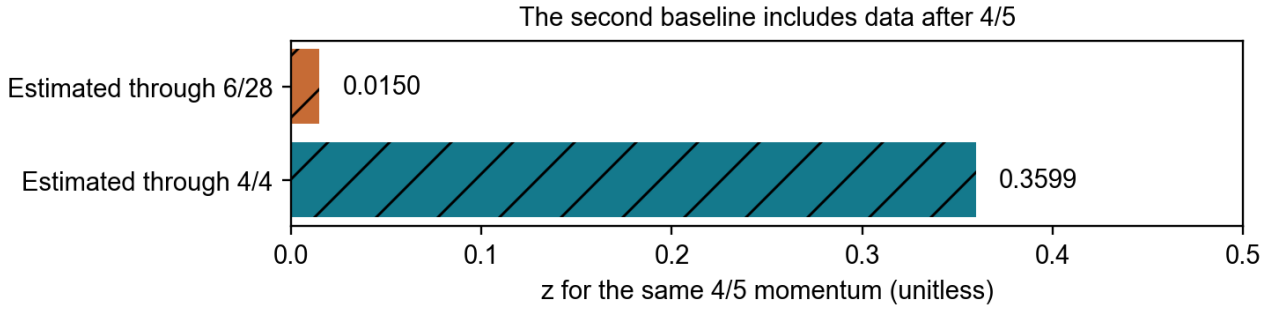
Changing only the estimation range for mean and standard deviation changes the same input's standardized value.

Keep the 4/5 momentum 0.0043829 and z=(m-μ)/σ̂. Compare estimates through 4/4 and through 6/28. A 4/5 decision cannot use a baseline estimated from later data.

Predict: Do the two z values differ because the 4/5 price changed or because μ and σ̂ changed?

```
full_mu = momentum3.mean()
full_scale = momentum3.std(ddof=1)
z_full = (apr5-full_mu)/full_scale
pd.DataFrame({'Baseline through':["4/4", "6/28"],
    'Mean':[mu, full_mu], 'Standard deviation':[scale, full_scale],
    "z":[z_earlier, z_full]}).round(7)
```

Baseline through	Mean	Standard deviation	z
4/4	-0.0038024	0.0227433	0.3598996
6/28	0.0039716	0.0274261	0.0149971



0.3599 and 0.0150 differ because of the estimation period. Smaller does not mean more accurate; the latter was unavailable at decision time.

Discussion

If μ and σ̂ use data through 6/28, can z=0.0150 be claimed as available at the 4/5 close?

Sources: [6], [14].

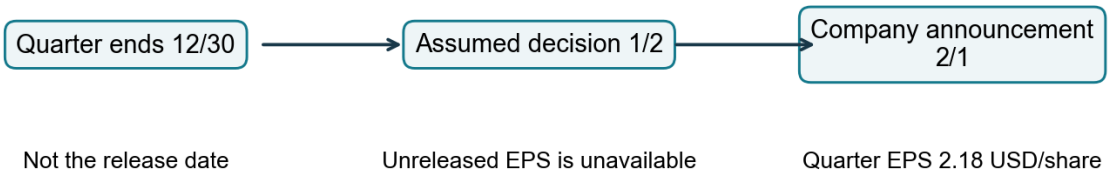
ISSUE 31

The quarter ended in December (month 12); is EPS 2.18 available in January (month 1)?

The financial reporting period is not the date its numbers became public.

EPS is earnings per share (USD/share); diluted EPS accounts for potential rights that increase shares. It is neither stock price nor cash dividend. Apple's quarter ended on 2023/12/30; EPS 2.18 was announced on 2024/2/1.

Observation date	Event	Is this quarter's EPS known?
2023/12/30	Reporting period ends	Period-end date alone does not establish availability
2024/1/2	Assumed decision date	The announcement is not yet released
2024/2/1	Company releases results	2.18 USD/share announced



SEC filings and company announcements have separate records. For the chosen feature source, check when the value was actually released, not just its period-end label.

A database retrospectively labeling 2.18 with 12/30 does not make it available on 12/30. Daily data also do not establish exact receipt time on the announcement date.

Discussion

If a model uses this quarter's EPS 2.18 on 1/2, which date evidence shows later information was used?

Sources: [7], [22], [25], [28].

ISSUE 32

Why can price fall despite 16% annual EPS growth?

Earnings growth compares two fiscal quarters; price return compares two trading days. They measure different quantities.

Apple's diluted EPS is 1.88 in the corresponding prior-year quarter and 2.18 USD/share this quarter. The quarters have 14 and 13 weeks respectively; this quarter is one week shorter. The annual growth rate therefore compares unequal periods. EPS growth is not received investment income or a guaranteed next-day price increase.

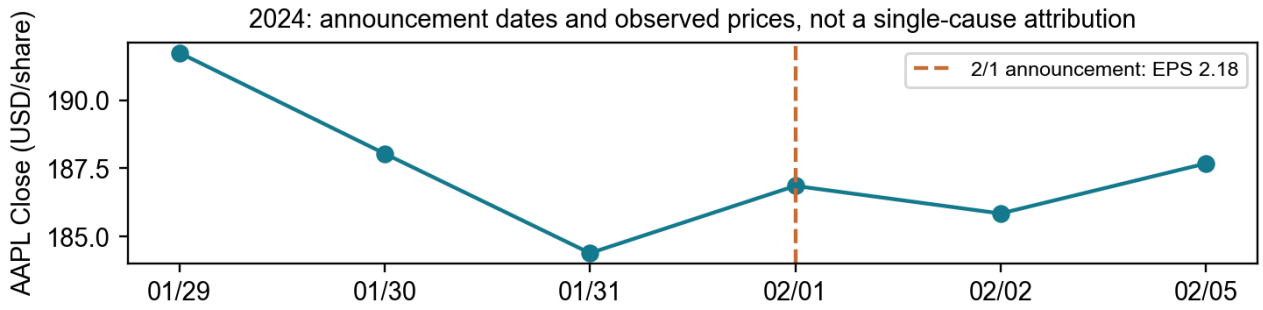
If released after the 2/1 close, the 2/2 Close is the first regular-session close afterward. The 2/1 to 2/2 return compared here still contains other information.

$$\text{EPS growth} = \frac{2.18}{1.88} - 1 \approx 15.96\%, \qquad r_{\text{Feb2}} = \frac{185.85}{186.86} - 1 \approx -0.54\%$$

Predict: Does positive EPS growth require a positive next-day price return?

```
eps_previous, eps_current = 1.88, 2.18  # Apple financial statements
eps_growth = eps_current/eps_previous-1
price_after_announcement = close["2024-02-02"]/close["2024-02-01"]-1
print(f"Annual EPS growth:{eps_growth:+.2%}")
print(f"2/1 to 2/2 Close return:{price_after_announcement:+.2%}")
```

Annual EPS growth:+15.96% 2/1 to 2/2 Close return:-0.54%



Close is 186.86 on 2/1 and 185.85 USD/share on 2/2. EPS growth 15.96% and price return -0.54% concern different periods.

This event alone cannot establish market expectations, isolate other news, or prove financial reports unimportant.

Discussion

Under the assumed release after the 2/1 close, why not use the 1/31 to 2/1 Close return to assess the market's response?

Sources: [22], [25], [28].

ISSUE 33

On 2/1, is the later revised GDP growth of 3.2% usable?

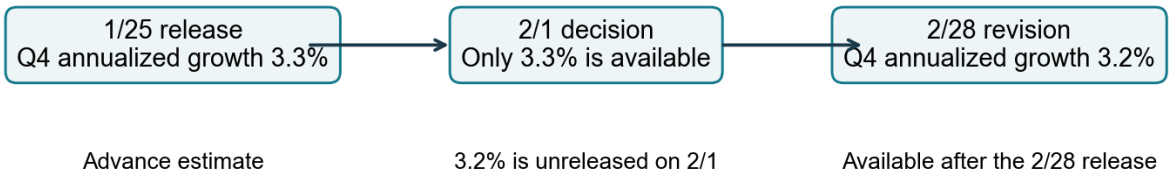
A quarter's estimate may be revised; a historical decision must use the version already released at that time.

GDP means gross domestic product; real GDP removes price changes. The US Bureau of Economic Analysis (BEA) released 2023 Q4 real GDP growth of 3.3% on 1/25, revised to 3.2% on 2/28. These are seasonally adjusted annual rates, not GDP levels, single-quarter growth or forecasts. Both releases were at 08:30 New York time.

Predict: Will 2/1 and 3/1 match the same version or different versions?

```
releases = pd.DataFrame({'Release_date':pd.to_datetime(
    ["2024-01-25","2024-02-28"]),'Annualized real GDP growth (%)':[3.3,3.2]})
decisions = pd.DataFrame({'Decision_date':pd.to_datetime(
    ["2024-02-01","2024-03-01"])})
# Match each decision date to the latest release no later than that date
matched = pd.merge_asof(decisions,releases,
    left_on='Decision_date',right_on='Release_date',direction="backward")
matched.assign(Decision_date=matched['Decision_date'].dt.strftime("%Y-%m-%d"),
    Release_date=matched['Release_date'].dt.strftime("%Y-%m-%d"))
```

Decision_date	Release_date	Annualized real GDP growth (%)
2024-02-01	2024-01-25	3.3
2024-03-01	2024-02-28	3.2



3.2-3.3=-0.1 percentage points is a revision for the same quarter, not economic growth from month 1 to month 2.

The annual rate compounds the quarter's change as if it continued for four quarters; it does not mean GDP already rose 3.3% in one quarter.

Discussion

Using 3.2% on 2/1 brings forward a value released on which date?

Sources: [8], [9], [16].

ISSUE 34

Which GDP version belongs in each daily feature row?

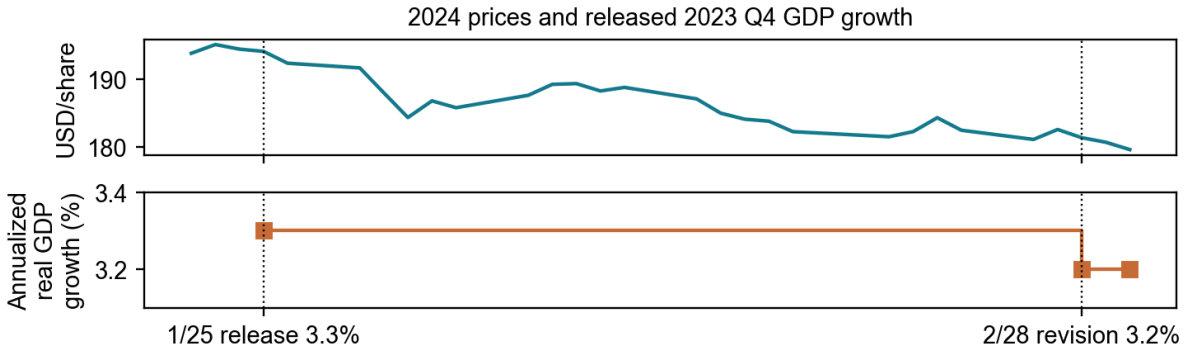
Quarterly GDP estimates are released weeks apart, while stock prices are observed daily. Each daily feature row can use only the latest version released before that day's close.

Use releases from Issue 33: 2023 Q4 annualized real GDP growth, released as 3.3% on 1/25 and revised to 3.2% on 2/28. Both were released at 08:30 New York time, before regular trading. This is a growth rate, not a GDP level.

Predict: Should the GDP row for 2/27 contain 3.3% or 3.2%?

```
days = pd.DataFrame({'Observation date':pd.to_datetime(
    ["2024-01-25","2024-01-26","2024-02-27","2024-02-28"])})
# Carry the latest version no later than each trading day until a new release
daily_gdp = pd.merge_asof(days,releases,left_on='Observation date',
    right_on='Release_date',direction="backward")
day_labels = daily_gdp['Observation date'].dt.strftime("%Y-%m-%d")
pd.DataFrame({'Observation date':day_labels.values,
    'AAPL closing price':close.loc[day_labels].round(2).values,
    'Available annualized growth (%)':daily_gdp['Annualized real GDP growth (%)'].values})
```

Observation date	AAPL closing price	Available annualized growth (%)
2024-01-25	194.17	3.3
2024-01-26	192.42	3.3
2024-02-27	182.63	3.3
2024-02-28	181.42	3.2



AAPL closes at 194.17 on 1/25 and 181.42 on 2/28. GDP is 3.3% in each row from 1/25 through 2/27, then 3.2% from 2/28. A flat line means no new version, not an unchanging economy.

Carry the quarterly value until the next release. Do not backfill 3.2% to 2/27 or earlier, or attribute this price decline solely to the GDP revision.

Discussion

If the 2/28 revision were released after the US stock-market close, which GDP version belongs in a 2/28 close-time decision? What would this page's date-only merge_asof insert?

Sources: [8], [9], [16].

References and Reproduction

The following sources support calculation definitions, common uses and data dates.

[1] yfinance: historical prices

[2] Nasdaq: regular trading hours

[3] pandas: rolling windows

[4] pandas: exponential weighting

[5] pandas: shifting rows

[6] scikit-learn: preventing future-data leakage

[7] SEC: Apple 10-Q filing record

[8] BEA: 2023 Q4 GDP advance estimate

[9] BEA: second GDP estimate for the same quarter

[10] Fidelity: SMA

[11] Fidelity: EMA

[12] Lo, Mamaysky and Wang: technical analysis research

[13] Fidelity: rate of price change

[14] NIST: standard deviation

[15] Nasdaq: trading volume

[16] pandas: matching by an earlier release date

[17] Yahoo Finance: AAPL historical prices

[18] Fidelity: MACD

[19] StockCharts: ATR calculation and initialization

[20] StockCharts: RSI calculation and use

[21] Fidelity: TR and ATR

[22] Apple: earnings announcement on 2024/2/1

[23] TradingView: MACD histogram

[24] Fidelity: technical analysis and common uses

[25] Apple: 2024 Q1 financial statements, page 1

[26] Fidelity: historical volatility and risk

[27] Investor.gov: stop triggers do not guarantee fill prices

[28] SEC: EPS

Reproduce with retained Yahoo/yfinance data, without a new download. Main AAPL dates are 2024/1/2-6/28; SMA200 and financial-report figures use history from 2023, with matching overlapping Closes. Indicator calculations can be checked; this does not verify historical data receipt or actual fills.